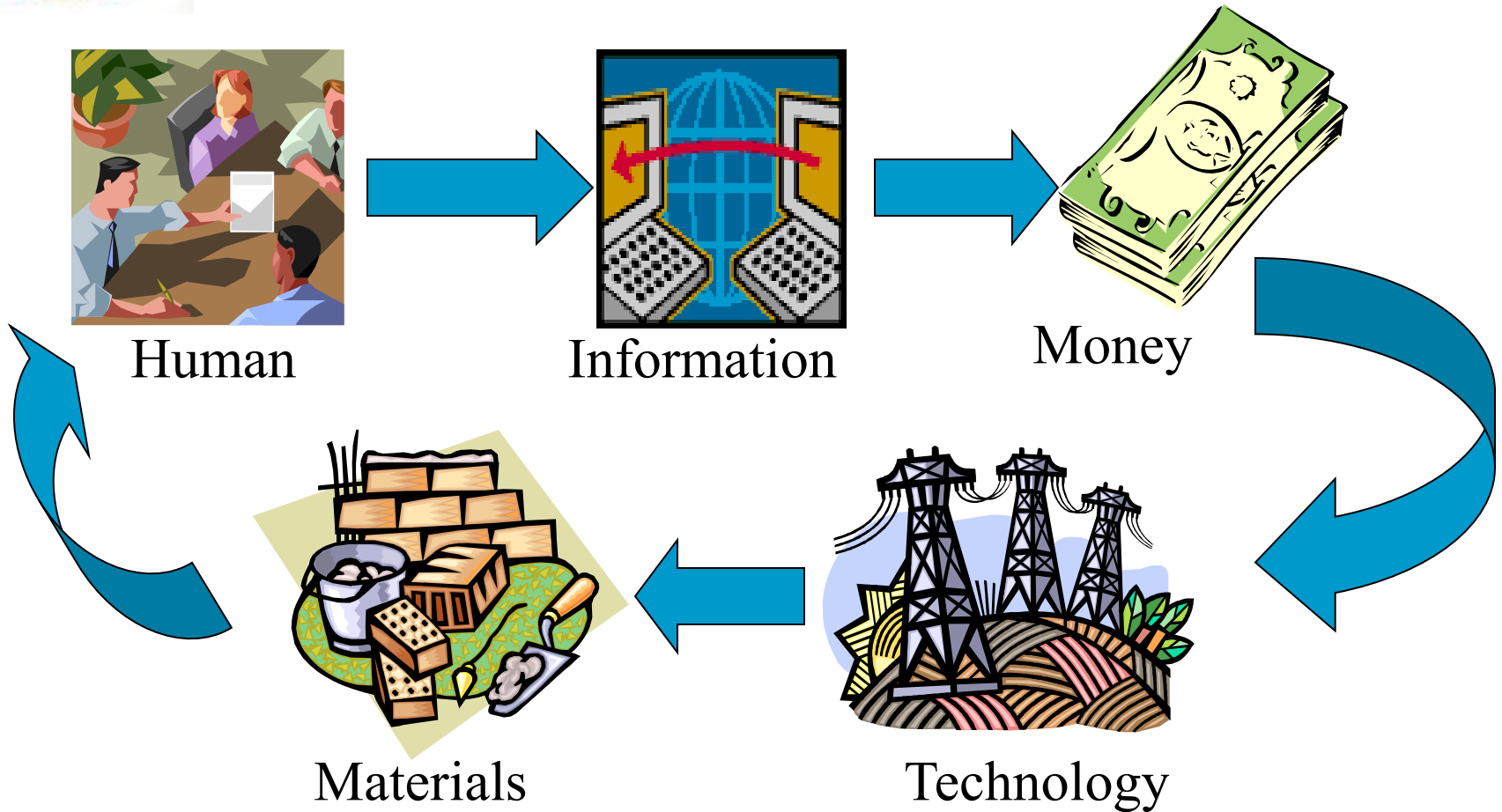




Industrial Engineering integrates:



to produce goods and services efficiently and optimally.



Management Functions in an Industrial Organization

– Planning

- **Defining goals, establishing strategies to achieve goals, developing plans to integrate and coordinate activities**

– Organizing

- Arranging work to accomplish organizational goals

– Leading

- Working with and through people to accomplish goals

– Controlling

- Monitoring, comparing, and correcting the work



Operations Research, defined

- Operations Research (OR) is defined as the application of *scientific methods*, tools and techniques by an *interdisciplinary team* to problems involving the operations of complex systems of people and machines. It aims to provide the *optimum or close to optimum solutions* to the problems of people who manage and control the operations.
- Operations Research (OR) is the discipline of applying advanced analytical methods to help ***make better decisions***.
- Historical Background
- Theory vs Applications (To prove or not to prove, that is the question...)



Analytics, defined

- Analytics is the application of scientific and mathematical methods to the study and analysis of problems involving complex systems. There are three distinct types of analytics.
- *Descriptive Analytics* gives insight into past events, using historical data.
- *Predictive Analytics* provides insight on what will happen in the future.
- ***Prescriptive Analytics*** helps with ***decision-making*** by providing actionable advice.



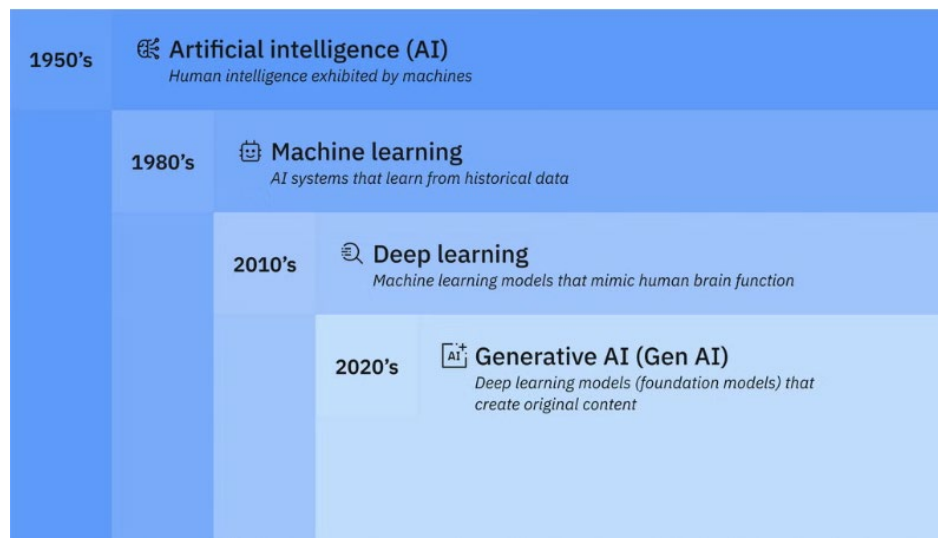
Operations Research (OR) vs Analytics

- Though similar in definition, and there are instances of overlap, Operations Research and Analytics are actually two unique but related fields.
- OR employs highly developed methods using advanced tools and techniques to provide analytical power. This is something that no ordinary software or spreadsheet can deliver out of the box. OR draws upon the latest analytical technologies including simulation, optimization, probability, and statistics.
- Analytics helps realize business objectives by *analyzing data* to create *predictive models* for forecasting and **optimizing** business processes for enhanced performance.



Artificial Intelligence (AI), defined

- Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, **decision making**, creativity and autonomy.
- A simple way to think about AI is as a series of nested or derivative concepts that have emerged over more than 70 years:



- Now in the mid 2020's, Agentic AI describes AI systems that are designed to **autonomously make decisions** and act, with the ability to pursue complex goals with limited supervision. It brings together the flexible characteristics of large language models (LLMs) with the accuracy of traditional programming.



OR vs AI

- The parallel evolution of OR and AI is deeply intertwined with the development of modern computing, although each field emerged from distinct intellectual traditions and followed distinct trajectories.
- Both the Artificial Intelligence (AI) and the Operations Research (OR) communities are interested in developing techniques for solving hard combinatorial problems.
- Both OR and AI have been influenced by the rise of data and computing capabilities, which continue to progress at tremendous speeds. Throughout these transformative years, the distinctions between OR's structured mathematical models and AI's heuristic flexibility have persisted in many cases.
- Although OR excels at well-defined problems amenable to mathematical formulation, the field's enduring strength has been its problem-driven, impact-oriented perspective that prioritizes practical applications over theoretical elegance alone.
- This emphasis aligns well with AI's development of tools for tackling ill-structured, knowledge-rich domains that defied precise mathematical representation. There are thus natural opportunities for collaboration between these groups, although these opportunities have not been fully realized.



Major Areas of Specialization in OR

- Linear Programming
- Integer Programming
- Game Theory
- Dynamic Programming
- Simulation
- Stochastic Processes
- Nonlinear Programming
- Heuristics



Major Applications of OR

- Supply Chain – sourcing decisions, make vs. buy decisions
- Manufacturing - production mix, production planning and scheduling, scheduling, line balancing, queueing
- Warehousing, Transportation & Logistics – multi-stage inventory decisions, transportation, transshipment, traveling salesman problem
- Financial Planning – capital budgeting, return with risk analyses, own vs. lease decisions, replacement policies



Major Applications of OR

- Investment Analysis – investment portfolio management
- Sales and Marketing – deployment of sales force, allocation of marketing budget, bidding strategies
- Facilities and Equipment Planning – plant location, equipment layout, capacity planning, vehicle fleet planning, equipment maintenance/replacement decisions
- Manpower Planning – manpower leveling, work assignment and deployment
- Material Planning – feed mix formulation, ingredient composition, trim loss problem
- Gaming – corporate rivalry strategies, sports strategies, war strategies, bids and auctions



OR Methodology

1. Definition of the Problem
- 2. Construction of the Model (Mathematical Model) ***
- 3. Solution of the Model ***
4. Validation of the Model
5. Implementation of the Solution

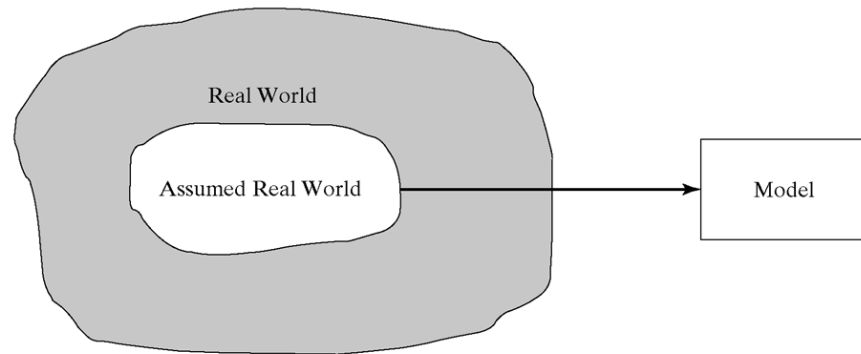


Figure 1.1
Levels of abstraction in model development.